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## **PELVIC MUSCLES** **AND MUSCLES OF THE GLUTEAL REGION**

### **PLAN**

#### **PELVIC MUSCLES**

- I. INTRODUCTION**
- II. DESCRIPTIVE ANATOMY**
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  2. Iliacus Muscle
  3. Psoas Minor Muscle

#### **MUSCLES OF THE GLUTEAL REGION**

- I. INTRODUCTION**
- II. DESCRIPTIVE ANATOMY**
  - A. Gluteal Muscles and Tensor Fasciae Latae
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  - B. Pelvi-Trochanteric Muscles
    - 1) Piriformis
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    - 3) Superior and Inferior Gemelli
    - 4) Obturator Externus
    - 5) Quadratus Femoris

### **OBJECTIVES**

- Know the muscular insertions.
- Know the actions of the muscles.
- Know their innervations.

## **PELVIC MUSCLES**

### **I. INTRODUCTION:**

The pelvic muscles are represented by two muscles: the psoas major and the iliacus muscle, grouped together under the name iliopsoas muscle. An inconstant muscle, the psoas minor, may also be present.

### **II. DESCRIPTIVE ANATOMY**

#### **1. Psoas Major Muscle**

- Origin:
  - Lateral surface of the vertebral bodies from T12 to L5 by fibrous arches
  - corresponding intervertebral discs
  - Inferior border of the costiform processes.
- Body: Fusiform muscle.
- Insertion: Tendon onto the summit of the lesser trochanter.
- Action: Flexion of the thigh on the pelvis; lateral rotation of the thigh.
- Innervation: Lumbar plexus.

#### **2. Iliacus Muscle**

- Origin:
  - Inner lip of the iliac crest
  - Iliac fossa.
- Body: Thick, broad, triangular muscle.
- Insertion: Lesser trochanter.
- Action: Flexion of the thigh on the pelvis; lateral rotation of the thigh.
- Innervation: Femoral nerve.

#### **3. Psoas Minor Muscle**

- Origin: Lateral surface of the bodies of T12 and L1 vertebrae and the intervening intervertebral disc.
- Body: Fusiform muscle.
- Insertion: Arcuate line.
- Action: Accessory flexor of the pelvis.
- Innervation: Branch of the lumbar plexus.

## **MUSCLES OF THE GLUTEAL REGION**

### **I. INTRODUCTION**

The muscles of the gluteal region are divided into two groups: the gluteal muscles and tensor fasciae latae; and the pelvi-trochanteric muscles.

### **II. DESCRIPTIVE ANATOMY**

#### **A. Gluteal Muscles and Tensor Fasciae Latae**

Arranged in three plans:

##### ❖ Deep layer:

##### **1) Gluteus Minimus:**

- Origin: Lateral surface of the ilium anterior to the anterior gluteal line.
- Body: thick, flattened and triangular muscle.
- Insertion: Anterior border of the greater trochanter.
- Action: Abduction of the thigh; medial rotation of the thigh.
- Innervation: Superior gluteal nerve (sacral plexus).

##### ❖ Medium layer

##### **2) Gluteus Medius:**

- Origin: Lateral surface of the ilium between the iliac crest and the anterior and posterior gluteal lines.
- Body: thick, flattened and triangular muscle
- Insertion: Lateral surface of the greater trochanter.
- Action: Abductor of the thigh; medial rotation (anterior fibers) and lateral rotation (posterior fibers).
- Innervation: Superior gluteal nerve (sacral plexus).

##### ❖ Superficial layer:

##### **3) Gluteus Maximus**

- Origin:
  - Posterior quarter of the iliac crest
  - Lateral surface of the ilium posterior to the posterior gluteal line
  - Lateral sacral crest
  - Lateral borders of the sacrum and coccyx.
- Body: a very thick, broad muscle, the largest and most powerful muscle in the body, formed of two superficial and deep bundles.
- Insertion: Superficial fibers to the iliotibial tract; deep fibers to the gluteal tuberosity of the femur.
- Action: Extension and lateral rotation of the thigh; assists in adduction and abduction.
- Innervation: Inferior gluteal nerve (sacral plexus).

#### **4) Tensor Fasciae Latae:**

- Origin:
  - Lateral lip of the iliac crest
  - Anterior superior iliac spine
  - Gluteal fascia.
- Body: muscle length, tendinous at the bottom.
- Insertion: Upper part of the iliotibial tract.
- Action: Tenses the iliotibial tract; extends the leg; stabilizes the hip and knee joints.
- Innervation: Superior gluteal nerve (sacral plexus).

#### **B. Pelvi-Trochanteric Muscles:**

##### **1) Piriformis muscle:**

- Origin: Anterior surface of the 2nd, 3rd, and 4th sacral vertebrae.
- Body: flattened muscle in the shape of a triangular pyramid.
- Insertion: Superior surface of the greater trochanter.
- Action: Lateral rotation of the thigh.
- Innervation: Nerve to piriformis (sacral plexus).

##### **2) Obturator Internus:**

- Origin: Internal surface of the obturator foramen and obturator membrane.
- Body: flattened and triangular muscle
- Insertion: Medial surface of the greater trochanter.
- Action: Lateral rotation of the thigh.
- Innervation: Nerve to obturator internus (sacral plexus).

##### **3) Superior and Inferior Gemelli:**

- Origin:
  - Superior gemellus from the ischial spine
  - Inferior gemellus from the ischial tuberosity.
- Body: accessory fleshy bundles of the internal obturator.
- Insertion: Tendon of the obturator internus.
- Action: Lateral rotation of the thigh.
- Innervation:
  - Superior gemellus : nerve of the internal obturator and superior gemellus.
  - Inferior gemellus : nerve of the inferior gemellus and quadratus femoris.

##### **4) Obturator Externus:**

- Origin: External surface of the obturator membrane and margins of the obturator foramen.
- Body: flattened and triangular muscle.
- Insertion: Trochanteric fossa.
- Action: Lateral rotation of the thigh.

- Innervation: Obturator nerve (lumbar plexus).

### 5) **Quadratus Femoris:**

- Origin: Lateral border of the ischial tuberosity.
- Body: thick, flat and quadrilateral muscle.
- Insertion: Intertrochanteric crest of the femur.
- Action: Lateral rotation of the thigh.
- Innervation: Nerve to quadratus femoris.

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